

Three-Stage Bearing Health Classification Using Vibration Short-Time Fourier Transform and Temperature Trends in a Run-to-Failure Setting

Long Truong^{1†} and Phu Le Nguyen^{2*†}

¹Faculty of Engineering and Technology, FPT University, Ho Chi Minh City, Vietnam.

²Faculty of Engineering and Technology, Nguyen Tat Thanh University, Ho Chi Minh City, Vietnam.

*Corresponding author(s). E-mail(s): lpnguyen@ntt.edu.vn;

Contributing authors: longt5@fpt.edu.vn;

[†]These authors contributed equally to this work.

Abstract

In this work, the three-stage bearing health classification (Healthy, Degrading, Fault) under a run-to-failure setting is addressed using vibration short-time Fourier transform (STFT) representations and compact temperature trend descriptors. A lightweight multi-modal pipeline is presented in which per-frequency (across-time) normalization stabilizes spectrogram brightness and a late head fuses a minimal 6-D temperature descriptor with vibration features. Evaluation is conducted with a leakage-aware, time-to-failure (TTF)-aligned protocol and a two-phase workflow (stratified pre-training followed by temporal fine-tuning) to mitigate temporal leakage and reflect deployment. On a stratified development split, Macro-F1 is ≈ 0.7762 . Under leakage-aware temporal testing, strong late-life discrimination is achieved (Fault F1 = **0.9600** on 90–100% TTF), and full-lifecycle performance reaches Accuracy = **0.8915** and Macro-F1 = **0.8739** with majority vote, demonstrating clear gains over a vibration-only baseline.

Keywords: bearing health monitoring, run-to-failure, degradation stage classification, STFT spectrogram, vibration–temperature fusion, temporal split, condition-based maintenance

1 Introduction

This work targets two practical challenges in run-to-failure bearing monitoring: (i) reliable recognition of the *degrading* stage—often subtle and class-imbalanced—and (ii) *leakage-aware* evaluation that reflects deployment reality rather than optimistic window-level splits [1–3]. A lightweight multi-sensor approach is adopted in which vibration provides time–frequency evidence and temperature contributes complementary trends near degradation onset [4]. STFT-based CNN features are coupled with a compact temperature descriptor via late fusion, and validation is conducted using a contiguous, TTF-aligned protocol. To bridge development and deployment, a two-phase workflow is followed: file-wise stratified training for model selection, then temporal fine-tuning aligned with the TTF axis (train [0,60]%, val [60,70]%, test [70,100]%) to test generalization from earlier to later life while reducing overlapped-window leakage. **Contributions.** This paper makes the following contributions, focusing on what is new and practically useful:

- A compute-efficient multi-modal pipeline is proposed that fuses STFT-based two-axis vibration features with a minimal 6-D temperature trend/statistical descriptor via a late-fusion head, together with per-frequency (across-time) normalization to stabilize spectrogram brightness.
- A leakage-aware, TTF-aligned evaluation recipe with present-class scoring is employed to reflect deployment and avoid window-overlap leakage; temporal slices use file-wise mean-logit aggregation, while full-range evaluation additionally reports majority vote when it proves stronger.
- Targeted ablations (vibration-only vs. multi-modal, stratified initialization vs. scratch, temporal slice selection) are provided to explain when temperature adds value and how initialization affects temporal generalization.
- Empirically, strong performance is achieved: Macro-F1 ≈ 0.7762 on a 3-class stratified split; under leakage-aware temporal testing (present classes), **Early (70–90%): Accuracy** = 0.8846, **Degrading F1** = 0.9388 and **Late (90–100%): Accuracy** = 0.9231, **Fault F1** = 0.9600. Over the full range [0, 100]% with majority vote, Accuracy = 0.8915 and Macro-F1 = 0.8739.

Time–frequency CNNs remain strong baselines for bearing diagnosis, while multi-modal fusion can improve robustness near transition regions by combining vibration with slower thermal trends [4–13]. Related work also includes direct 1D-to-2D grayscale conversions of vibration signals; here STFT is preferred because it preserves explicit time–frequency structure with a compact preprocessing pipeline [8]. Unlike studies based on random or window-level splits, this paper evaluates the method under a leakage-aware, TTF-aligned protocol to better approximate deployment [2, 3].

2 Proposed Framework

2.1 Pipeline overview

A lightweight multi-modal pipeline is presented. Each 1.0-s window yields a 2-channel log-spectrogram from two-axis vibration and a compact 6-D descriptor from two-channel

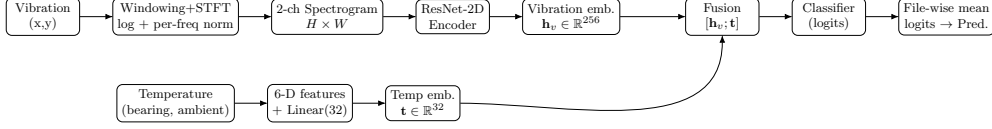


Fig. 1 Block diagram of the proposed multi-modal pipeline with late fusion and file-wise mean-logit aggregation.

temperature. A small 2D CNN encodes vibration, a linear layer embeds temperature, and a late-fusion classifier predicts the stage. Figure 1 summarizes the end-to-end flow and where fusion and file-wise aggregation occur.

2.2 Signal processing, model, and inference

Vibration (2 axes) and temperature (2 channels) are sampled synchronously and sliced by identical 1.0-s windows with 50% overlap, avoiding extra timestamp alignment. (If rates differ in deployment, a simple resampling/alignment step is required.)

Each window is sanitized (NaN/Inf $\rightarrow$ 0) and z-scored per channel. STFT log-magnitude spectrograms are computed with $n_fft = 2048$, $hop_length = 512$ (Hann), then resized to 160×160 . A per-frequency (across-time) normalization stabilizes spectral brightness; the two vibration axes are then stacked as channels to form a $(2 \times H \times W)$ input. In parallel, a compact 6-D temperature descriptor (mean, std, slope for each of two channels) is computed per window.

The vibration branch uses a compact ResNet-style 2D encoder [14]. The temperature descriptor is projected to a 32-D embedding and concatenated with the vibration embedding immediately before the linear classifier, i.e., standard late fusion. Training follows two phases: file-wise stratified learning for model selection, then temporal fine-tuning with the leakage-aware TTF split (train [0,60]%, val [60,70]%, test [70,100]%) to transfer from earlier to later life under chronology-respecting evaluation.

Unless stated otherwise, inference uses file-wise *mean-logit* aggregation: *pre-softmax* logits are averaged across windows from the same file before applying $\arg \max$. For completeness, majority vote across window-level predictions is also reported for the full-range evaluation, where it proves stronger than mean-logit. Reported metrics are Accuracy and per-class Precision/Recall/F1; Macro-F1 is averaged only over classes that are present in the evaluated slice. When a slice becomes single-class (e.g., very late life), only the present-class F1 is reported. Hyperparameters for both training phases are summarized in Table 2; dataset-specific details are given in Section 3.

3 Experimental Setup

3.1 Experimental Configuration

The run-to-failure bearing dataset provided by Jung et al. [1, 15] is used in this study. The dataset contains **one run** (RUNS=1) with **129 files** and two-axis vibration sampled at **25.6 kHz**. Preprocessing follows Section 2: 1.0-s windows with 50% overlap, STFT with $n_fft = 2048$ and $hop_length = 512$, 160×160 spectrograms, and a 6-D temperature descriptor from two channels.

Let $\text{TTF\%} \in [0, 1]$ denote the normalized time-to-failure percentage within the run. Throughout the paper, TTF percentages are reported on the closed interval $[0, 100]\%$, where 100% denotes the terminal observed state of the run. Three stages are defined along normalized lifetime; see Table 1 for the threshold ranges that delineate early/mid/late life and guide temporal splits.

Table 1 Health stages by TTF% (normalized lifetime).

Stage	TTF% range
Healthy	$[0, 0.60)$
Degrading	$[0.60, 0.90)$
Fault	$[0.90, 1.00]$

All experiments use a single NVIDIA RTX 2060 (6 GB) GPU with PyTorch 2.x, CUDA 12.x, seed 42, and mixed precision. The canonical stratified reference checkpoint is runs/logs_stft_strat/auto_r22/best.pt; temporal reruns use configs/best_temporal.yaml initialized from that checkpoint, and full-range evaluation uses configs/best_fullrange_eval.yaml. Temporal validation uses at most 50 windows per file.

3.2 Development setting: stratified split

A file-wise stratified split is adopted for development (0.6/0.2/0.2). Training hyperparameters and settings are summarized in Table 2.

Table 2 Training settings (dev vs. temporal fine-tuning).

	Stratified (dev)	Temporal (FT)
Optimizer	AdamW	AdamW
LR	2×10^{-4}	3×10^{-5}
Weight decay	10^{-3}	10^{-2}
Epochs	80	60
Schedule	–	cosine (OneCycle off)

3.3 Deployment-oriented setting: leakage-aware temporal split

A contiguous, TTF-aligned temporal split is adopted to avoid overlapped-window leakage and better match deployment: train $[0, 60]\%$, val $[60, 70]\%$, test $[70, 100]\%$. The split is illustrated in Figure 2; remaining settings follow Table 2.

4 Results

The file-wise stratified split is used only for model selection. The best development checkpoint (auto_r22) reaches **Macro-F1 = 0.7762**; Fault is already highly separable,

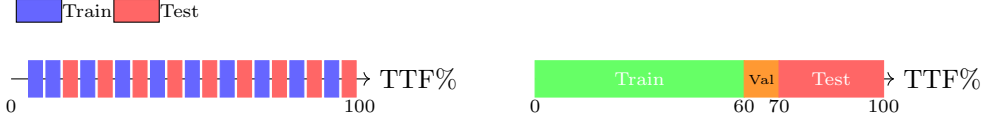


Fig. 2 Protocol comparison. Left: window-level random split (leakage risk). Right: contiguous TTF-aligned temporal split (leakage-aware).

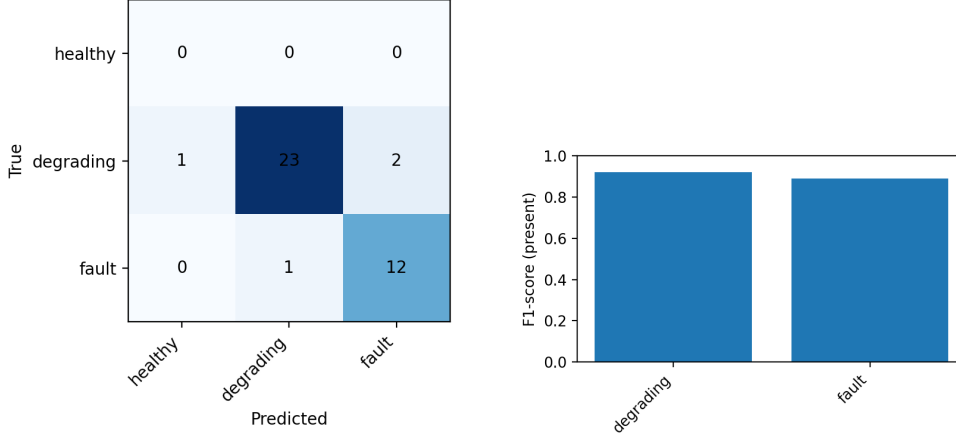


Fig. 3 Leakage-aware temporal evaluation on the deployment test slice [70,100]% TTF. Left: confusion matrix. Right: per-class F1 over present classes (Healthy absent).

while Degrading remains the hardest class. The main analysis therefore focuses on the leakage-aware temporal evaluation.

4.1 Temporal evaluation (deployment-oriented)

Under the temporal leakage-aware split, the deployment test slice [70,100]% contains only later-life behavior, and *Healthy samples do not appear*. Therefore, temporal metrics are interpreted over the *present classes* (Degrading and Fault). Using mean-logit aggregation on the canonical temporal configuration, the model reaches **Accuracy = 0.8974** and **Macro-F1_{present} = 0.9044** on the full [70,100]% slice, with **Degrading F1 = 0.9200** and **Fault F1 = 0.8889**. On sub-slices, performance remains strong: **Early (70–90%): Accuracy = 0.8846, Degrading F1 = 0.9388** and **Late (90–100%): Accuracy = 0.9231, Fault F1 = 0.9600**. These results suggest that the learned representation transfers from earlier life to later-life conditions; the Healthy–Degrading boundary around 60–70% TTF remains the main challenge when broader coverage is included.

Figure 3 summarizes temporal performance on the deployment test slice.

4.2 Baseline comparison

Table 3 reports file-wise performance of the three comparison baselines on both protocols. For the temporal protocol, Healthy is absent from the test slice [70,100]%; Macro-F1 is therefore computed over present classes (Degrading, Fault) only. Compared with the classical baseline using 8-D handcrafted vibration features and an SVM, the proposed multi-modal model is consistently stronger and the margin becomes substantial under the deployment-oriented temporal protocol. In the stratified setting, both methods reach the same Accuracy (0.7241), but the proposed model yields a higher Macro-F1 (0.7762 vs. 0.6190), indicating better class balance rather than gains concentrated in a single class. The difference is larger under temporal testing: the classical SVM reaches Accuracy = 0.6923 and Macro-F1_{present} = 0.5875, whereas the proposed model attains Accuracy = 0.8974 and Macro-F1_{present} = 0.9044. At the class level, the SVM achieves only partial Fault recognition (Fault F1 = 0.3750), while the proposed model separates both Degrading and Fault reliably (Degrading F1 = 0.9200, Fault F1 = 0.8889). The matched vibration-only CNN further shows that a deep architecture alone is insufficient here: under the same fair temporal protocol it reaches only Accuracy = 0.3333 and Macro-F1_{present} = 0.2500, effectively collapsing to a single-class prediction pattern and supporting the contribution of the multi-modal fusion design.

Table 3 Comparison of three baselines on stratified (development) and temporal [70,100]% (deployment) protocols, evaluated file-wise. Multi-modal stratified result is from the best auto-training checkpoint (round 22, n_fft=2048, 160×160). †Macro-F1 over present classes only (Healthy absent in temporal test).

Method	Stratified		Temporal [70,100]%	
	Acc	Macro-F1	Acc	Macro-F1 [†]
Classical vib (SVM, 8D)	0.7241	0.6190	0.6923	0.5875
Vibration-only CNN	0.1724	0.0980	0.3333	0.2500
Proposed multi-modal	0.7241	0.7762	0.8974	0.9044

For completeness, Figure 4 summarizes full-range behavior over [0,100]% TTF. Majority vote is the stronger file-wise aggregator (Accuracy = 0.8915, Macro-F1 = 0.8739) than mean-logit (Accuracy = 0.7752, Macro-F1 = 0.8006), and the remaining errors are concentrated mainly around the Healthy–Degrading transition.

4.3 Failure case analysis with broader temporal coverage

When evaluation expands to cover a broader range that includes more ambiguous mid-life samples, metrics can shift notably: Degrading is more frequently confused with Healthy and, depending on the slice, Healthy may have zero support. This highlights sensitivity near the Healthy–Degrading boundary and motivates stronger supervision and longer temporal context, as well as multi-sensor cues that remain informative near transition regions.

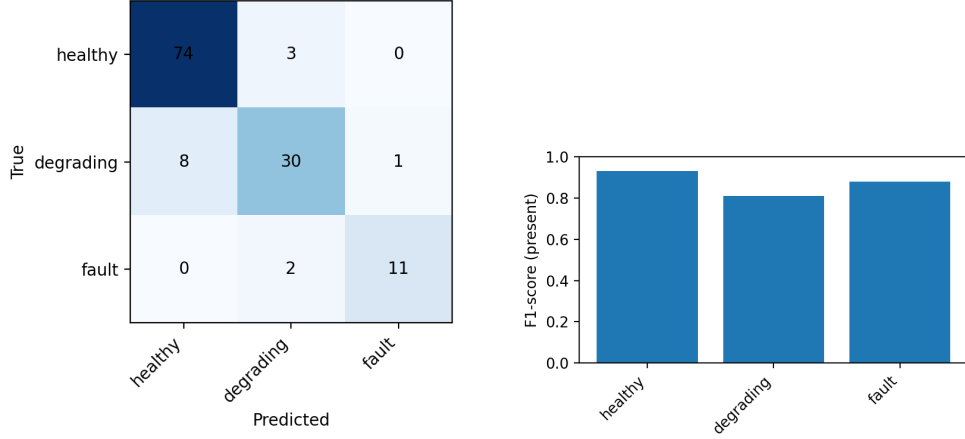


Fig. 4 Full-range temporal evaluation $[0, 100]\%$ TTF under file-wise majority-vote aggregation. Left: confusion matrix. Right: per-class F1.

5 Ablation and Discussion

5.1 Ablation Studies

Table 4 consolidates ablations across temporal slices using matched deep-model variants; the classical SVM comparison is kept separately in Table 3 to avoid mixing slice-wise exploratory runs with the canonical paper baseline. In particular, on a representative late-life slice $[85, 100]\%$, the multi-modal model (vibration+temperature) attains Macro-F1 = 0.9467 versus 0.2593 for vibration-only, with Fault F1 improving from 0.0000 to 0.9600, underscoring the benefit of multi-sensor fusion near transition and late stages. Key takeaways:

- **Modality effect.** On the matched temporal protocol, the vibration-only model becomes unstable: at $[70, 100]\%$ it collapses to a single-class solution (Macro-F1_{present} = 0.2500), and at $[85, 100]\%$ it fails to detect Fault (F1 = 0.0000), while the multi-modal model remains strong (Macro-F1_{present} = 0.9044 at $[70, 100]\%$ and 0.9467 at $[85, 100]\%$).
- **Initialization.** Initializing temporal fine-tuning from the best stratified checkpoint outperforms training from scratch (same coverage).
- **Slice sensitivity.** Broader coverage $[70, 100]\%$ reduces Degrading F1 due to the Healthy–Degrading boundary; very-late $[90, 100]\%$ can become single-class (Fault).
- **Variants.** A higher-resolution setting (4096/1024, 224×224) underperforms the final configuration on the late slice.

Table 4 Consolidated ablation summary across temporal slices (present-label metrics where applicable). *Note:* — indicates metrics not applicable due to absent labels or single-class slices.

Slice	Variant	Accuracy	Macro-F1 (present)	F1 Degrading	F1 Fault
[70, 90]%	Multi-modal (vib+temp)	0.8846	—	0.9388	—
[85, 100]%	Vib-only	0.3500	0.2593	0.5185	0.0000
[85, 100]%	Multi-modal (vib+temp)	0.9500	0.9467	0.9333	0.9600
[70, 100]%	Vib-only	0.3333	0.2500	0.0000	0.5000
[70, 100]%	Multi-modal (vib+temp)	0.8974	0.9044	0.9200	0.8889
[90, 100]%	Multi-modal (vib+temp)	0.9231	—	—	0.9600
[90, 100]%	High-res (4096/1024, 224×224)	0.8462	—	—	0.9167

Why temperature helps late life.

Temperature aggregates effects of load and friction; in late life (90–100% TTF) these trends become distinctive even when vibration patterns remain non-stationary. The multi-modal model leverages complementary cues, improving Fault recognition (Fault F1 = 0.9600) without sacrificing Degrading performance on earlier slices.

Why Healthy–Degrading is hard.

The transition around 60–70% TTF is gradual and class support is limited, yielding overlapping distributions. This manifests as lower Degrading F1 on broad slices that include earlier life. Sequence-aware context or stronger supervision near the boundary may further mitigate this issue.

Deployment considerations.

In practical monitoring, alarm thresholds typically target very-late life (e.g., > 90% TTF). The evaluation isolates early/late sub-slices and reports present-class metrics to reflect that some classes are absent, which stabilizes interpretation and aligns with decision points used in operations.

6 Conclusion and Future Work

This paper shows that a lightweight STFT+temperature late-fusion model remains effective under a leakage-aware, TTF-aligned evaluation protocol, with temperature becoming especially informative in very late life. The main residual weakness is the gradual Healthy–Degrading transition around 60–70% TTF, which also drives most errors under broader temporal coverage.

The current study is limited by single-run evaluation and empirically chosen stage thresholds. Future work will prioritize multi-run validation and stronger temporal modeling around the transition region.

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Declarations

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Competing interests. The authors declare that they have no competing interests.

Ethics approval and consent to participate. Not applicable.

Consent for publication. Not applicable.

Data availability. The dataset used in this study is publicly available [1, 15].

Materials availability. Not applicable.

Code availability. An anonymized code repository is provided to reviewers; code and configurations will be released publicly upon acceptance.

Author contributions. Long Truong and Phu Le Nguyen contributed equally to this work. Both authors contributed to conceptualization, methodology, experiments, and writing.

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